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What does 'qualifying text' mean?
Our model only processes qualifying text in the form of long-form writing. Long-form writing means individual sentences contained in paragraphs that make up a longer piece of written work, such as an essay, a dissertation, or an article, etc. Qualifying text that has been determined to be likely AI-generated will be highlighted in cyan in the submission, and likely AI-generated and then likely AI-paraphrased will be highlighted purple.

Non-qualifying text, such as bullet points, annotated bibliographies, etc., will not be processed and can create disparity between the submission highlights and the percentage shown.

A Routing Protocol for LoRa Mesh Networks

Given the limitations of LoRa gateways in poor internet connections this paper introduces a multi-hop system by adding repeater to extend range by combining the design of HWMP (Hybrid Wireless Mesh Protocol) and streamlined header information from AODV (Ad hoc On-demand Distance Vector). The authors of this paper survey on existing routing protocols for reliable communication and suggest that instead of interfering with existing LoRa implementations, apply the tunneling principle on it, which specifies packet formats and maintains routing and path information in the gateways using Routing Table and Device Table. This paper provides a detailed explanation on the methods of operations for the proposed system, its trade-offs and ways to mitigate them. Furthermore, feasibility of the system is demonstrated through quantitative comparisons of Rout Construct Time with artificial delays and field experimentation.

Beyond the Star of Stars: an Introduction to Multi Hop and Mesh for LoRa and LoRaWAN Pervasive Computing 4th submission

In addition to outlining a flexible structure for LoRa implementations using multi-hop mesh networks, this paper classifies the existing multi-hop solutions. Additionally, it identifies performance indicators (e.g. throughput, latency, scalability and energy efficiency). The authors recognize existing LPWAN systems as successful for IoT devices for long-distance communication; however, to address cases where the default system doesn't meet requirements, they consider three example scenarios (extending coverage, ensuring fault tolerance and enabling infrastructure-less deployment) and examine how the mesh solution can satisfy them.

Factors Influencing LoRa Communication in IoT Deployment Overview and Experience Analysis

This is a literature review on factors affecting signal quality and disturbance in LoRa communication in a “Smart Village” setup. This study uses RSSI (Received Signal Strength Indicator) as key comparison parameter with the help of PCA (Principal Component Analysis) to analyze the impacts of the diminishing factors on signal performance (e.g. building, vegetation, temperature, and mobility). The authors demonstrate correlation between these factors and signal propagation quality. Obstruction of the Fresnel Zone is identified as a major factor affecting signal stability. Additionally, the study presents ways to enhance quality, such as machine learning methods for optimal positioning, K-Nearest Neighbors (KNN) algorithm, Hitchcock Bird-Inspired (HCBI) algorithm and antenna positioning.

LoRa Performance in Industrial Environments: Analysis of Different Algorithms

This paper focuses on Adaptive Data Rate (ADR) algorithms using data driven analysis to understand how different algorithms perform in challenging environments with obstacles and interference. The study compares Packet Data Rate (PDR) and energy consumption across different algorithms, gateway placements and data traffic level using realistic propagation models derived from actual measurements. It also recommends specific ADR algorithms based on application needs. This paper contributes to the creation of a novel propagation model for worst-case scenarios and determination of characteristics that improve energy usage and packet delivery of the ADR algorithms.

PriNergy: A Priority-based Energy Efficient Routing Method for IoT Systems

Highlighting key challenges in large-scale applications, this paper emphasizes the need for an energy-efficient routing protocol to avoid data congestion and overhead and introduces PriNergy, based on the routing protocol for low-power and lossy networks (RPL) models. The protocol determines routing based on content priority and utilizes timing patterns in the network slots to send data, considering network traffic, audio, and image data. This routing method reduces mesh overhead, end-to-end delay and node energy consumption while optimizing packet transmission speed, outperforming Quality of Service RPL (QRPL). The Destination-Oriented Directed Acyclic Graph Information Object (DODAGIO) beacon process maintains routing topology, which allows nodes to reduce control message transmissions while preserving network stability. Time Division Multiple Access (TDMA) is used to modify frame values and synchronize time slots on the transmission channel. This protocol also employs Cyclic Redundancy Check (CRC) and other methods for frame error detection and correction.

SOHO network using Cisco packet tracer and LORa

This paper outlines the layout and implementation of a Small Office Home Office (SOHO) network using Cisco Packet Tracer for initial simulation and LoRa technology for physical wireless communication. The project focuses on the creation of inter-departmental communication between an administration and two other departments. It indicates the software requirements such as Cisco Packet Tracer and Arduino IDE, along with hardware elements like LoRa modules, Arduino Nano, and ESP32. The paper clarifies a process for setting up both the simulated and physical networks, demonstrating the manner in which messages and data packets are sent and displayed. Finally, the authors claim that this approach skillfully augments wireless communication within a small business setting.

LoRa Modulation: Frequency Shift Chirp Modulation (FSCM)

This 2017 research paper is the first academic work that gives a comprehensive mathematical description of LoRa Modulation (FSCM) and demodulation processes which has previously been patented. The authors detail the orthogonality of FSCM waveform, and formulates how the modulated signal (chirp waveform) varies based on initial frequency shift. They also derive an optimum receiver for LoRa modulation that uses a low-complexity demodulation process, utilizing Fast Fourier Transform (FFT). Finally, it compares the performance of FSCM against Frequency Shift Keying (FSK) modulation, demonstrating FSCM's superiority in frequency selective channels compared to Additive White Gaussian Noise (AWGN) channels.

Chirp Spread Spectrum-Based Waveform Design and Detection Mechanisms for LPWAN-Based IoT: A Survey

This comprehensive survey explores a variety of Chirp Spread Spectrum (CSS) based physical (PHY) layer alternatives to LoRa for Low Power Wide Area Network (LPWAN) Internet of Things (IoT) applications, while in-depth examining waveform designs that seek to enhance the data transmission of LoRa. The research classifies and examines fifteen alternatives based on their waveform design and groups them into single chirps (SC), multiple chirps (MC), and multiple chirps with index modulation (MC-IM) categories. The study examines the transceiver architectures, proposes detection mechanisms when they are not reported, and compares performance metrics like energy efficiency (EE), spectral efficiency (SE), and bit-error-rate (BER) for varying conditions. Finally, the paper highlights the advantages and disadvantages of each design approach, offering insights and identifying open research issues within the field.

Optimizing the Performance of Pure ALOHA for LoRa-Based ESL

This article details research focused on optimizing LoRa-based Electronic Shelf Label (ESL) performance, particularly concerning the Pure ALOHA communication protocol. The authors introduce and test an improved ALOHA mechanism that leverages orthogonal combinations of spreading factor (SF) and bandwidth (BW), alongside a K-means machine learning algorithm (MLA) for efficient end node (EN) arrangement. This approach aims to minimize network saturation and enhance successful data transmission, ultimately addressing the limitations of traditional ALOHA in LoRa networks for industrial applications like ESLs. The study concludes that while increasing gateways generally improves performance with MLA, there's a trade-off between bandwidth, energy consumption, and network complexity.

An Adaptive Spreading Factor Selection Scheme for a Single Channel LoRa Modem

This paper explains an Adaptive Spreading Factor Selection (ASFS) scheme for enhancing the throughput of LoRa (Long Range) networks, particularly in multi-hop setups based on more affordable single-channel modules. The authors propose the ASFS scheme to allow LoRa devices to automatically determine and synchronize their optimal spreading factor (SF), a crucial parameter that determines data rate and communication range, without using explicit packet transmissions. The paper explains the mechanisms by which this is achieved, including iterative SF testing and an improved SF selection algorithm, ultimately demonstrating through experimental evaluations that ASFS significantly improves end-to-end throughput compared to traditional LoRa systems, especially in multi-hop networks. This work aims to resolve the limitations of standard LoRaWAN's scalability and cost-effectiveness in complicated network deployments.